

Supporting Information for

Identifying Solar and Heliospheric Fronts With Conservation Laws and the MHD Riemann Problem

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Contents: Movies S1–S4 and an accompanying animation package containing the rendering script, retained inputs, and provenance records. The movies accompany the solar-image panels in Figure 1 and the SUVI and AIA comparisons in Figures 6 and 7 of the article.

All movies show observed images with fixed display scales within each sequence. The left panel shows brightness or radiance; the right panel shows a difference image. Red and blue indicate positive and negative changes. Playback is accelerated, and the printed UT timestamps give the observation times. Each exposure is held on screen; no image is interpolated between exposures. Previously calculated feature tracks and model positions are retained without refitting.

Movie S1. The 2010 June 13 AIA example (E11).

Sixteen AIA 193 Å exposures from 05:38:08.07 to 05:41:08.07 UT show the evolving off-limb structure, at approximately 12 s spacing. The right panel subtracts the reference at 05:35:08.07 UT. The retained displays use spatial Gaussian smoothing with a standard deviation of 1.20 arcsec and a difference scale of $\pm 12 \text{ DN s}^{-1}$. No temporal smoothing is applied. Turquoise and orange curves mark the retained inner and outer brightness-crest fits. A curve appears only in exposures for which that fit exists; it is not extrapolated into other frames. The curves describe image features. The conditional fast-shock test in the article concerns a prescribed plasma-state pair motivated by the observations discussed by Kozarev et al. (2011) and Ma et al. (2011).

Movie S2. The 2011 February 16 AIA example (E05).

Twenty-nine AIA 193 Å exposures from approximately 14:25:32.84 to 14:31:08.84 UT show the region used for the EIS comparison. The images are exposure-normalized and registered on a common grid. The first selected exposure supplies the base-difference reference; its difference panel is therefore zero. Yellow boxes show the sixteen retained alternatives for placement of the EIS sampling region. They do not trace one uniquely resolved front. The timestamps use the retained time offsets, anchored to the article exposure at 14:28:44.84 UT, and are rounded for display. The observational context is the analysis of Veronig et al. (2011). The two conditional state pairs in the article explore different interpretations of the incomplete plasma constraints.

Movie S3. The SUVI geometric test on 2017 September 10.

Five SUVI 195 Å exposures run from 15:54:24.75 to 15:58:24.76 UT. Their actual spacings are approximately 70, 50, 50, and 70 s. The left panel shows calibrated radiance on a logarithmic scale. The right panel subtracts the 15:30:24.69 UT reference after sampling on the retained common coordinate grid. It uses a fixed asinh color scale. No additional spatial or temporal smoothing is applied here.

Orange C marks the selected outer bright feature and pink E the selected inner edge. The dotted yellow outline marks the fixed trailing aperture B. At each time, the tested association places the P100 fast shock at C and its contact at E. One common shift and scale then determine the purple fast-rarefaction interval. This interval lies outside B from the second exposure onward. The labels show the model association being tested; they do not establish those wave types from image brightness. The movie visualizes the same geometric construction as the article. SUVI observations of this eruption were presented by Seaton et al. (2018).

Movie S4. The AIA comparison along G1 on 2017 September 10.

Five AIA 193 Å exposures run from 16:00:41.84 to 16:05:29.84 UT, at approximately 72 s spacing. The right panel subtracts each preceding exposure; the first uses 15:59:29.84 UT as its reference. These images are retained display levels used to compare positions and shapes. They are not used to infer density or temperature.

The gray curve is G1, an assumed great-circle path on a sphere of radius 696 Mm, projected into the image plane. The open turquoise circle marks the selected image maximum. Colored marks show the P100 wave positions: orange for the fast shock, pink for the contact, purple for the fast rarefaction, and green for the other model waves. Only the first fast-shock position is aligned with the selected maximum. All later positions follow from the fixed model speeds, the assumed initial model age of 120 s, and the same geometry. The movie shows outward feature motion together with the positions at which the other model waves can be sought. The contact and rarefaction remain predictions requiring independent plasma tests.

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The code archive provides `run.py`, `requirements.txt`, and all retained movie inputs. After installing the dependencies, run `python run.py animations`. Python and Matplotlib generate the annotated figures; FFmpeg encodes MP4 files. GIF copies and representative PNG/PDF frames are also generated. The README explains the inputs and command; `MANIFEST.json` records file checksums. The rendering code and numerical calculations are included in the release archived at <https://doi.org/10.5281/zenodo.23084162>.

AIA data are courtesy of NASA/SDO and the AIA science team. SUVI data are provided by NOAA/NCEI. References cited in these captions are listed in the article bibliography.